

Miłosz Szymański

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RESEARCH INTERESTS

My research interests include noncommutative ring theory, algebraic geometry, representation theory, algebraic topology, category theory, model theory, and the interactions between all these fields. I am also interested in geometrical and algebraic problems arising from mathematical physics.

EDUCATION

MMath Degree - University of Edinburgh

Sep. 2021 - Jul. 2026

Integrated Masters in Mathematics with Honours, First Class.

Listed below are some relevant courses I have taken, a full academic transcript is available upon request.

Algebraic Geometry

Galois Theory

Topics in Rings and Representation Theory

Representation Theory

Algebraic Topology

Differential Geometry

Category Theory

Group Theory

Axiomatic Set Theory

RESEARCH EXPERIENCE

Finite Dimensional Noetherian Filtrations

Sep. 2025 - Mar. 2026

For my dissertation, I studied notions of dimension generalised to noncommutative spaces, and how they can be used to answer classification questions. Specifically, I considered the classification of AS-regular algebras of global dimension three as detailed in *Artin&Schelter* 1987. Additionally, I used these ideas to examine counterexamples in *Stephenson&Zhang* 1997, and extend the work on counterexamples presented in *McConnell&Stafford* 1988, to the question sated in *McConnell&Robson* 1987, concerning finite-dimensional filtrations of algebras and the Noetherianity of their associated graded algebras. Supervised by Prof. Susan Sierra.

Matroid Applications in Phylogenetic Algebraic Geometry

Jul. 2025 - Sep. 2025

Studied the correspondence between phylogenetic group-based mixture models and their associated algebraic matroids, as well as the results concerning the generic identifiability of these models as presented in *Hollering&Sullivant* 2021. Based on the authors' code written in the Mathematica system, I wrote a novel implementation of the generic identifiability algorithm in the SageMath system. Collected my findings, including a complete exposition of the underlying matroid theory used in their results in a scientific report. Supervised by Dr Dimitra Kosta.

Intuitionistic and Many-Valued Logics

Sep. 2024 - Mar. 2025

Examined the results of Gödel and Jaśkowski which relate Intuitionistic Logic and Many-Valued Logics as part of a group research project, which was awarded an A grade. I was responsible for providing a detailed exposition of Mathematical Logic used in the statement and proofs of the results, as well as for setting the calendar for the project to ensure steady and timely progress. Towards the end of the project, we gave a well-received 30-min presentation to an audience of students and professors, in a mock-conference setting. Supervised by Prof. Antony Maciocia.

Phylogenetic Algebraic Geometry

Jun. 2024 - Aug. 2024

Conducted research into the applications of algebraic geometry in the study of phylogenetics. The linked report presents a summary of key techniques used in this field including notes on phylogenetic invariants, the discrete Fourier transform, group-based models, etc. as well as numerous explicit calculations, using the Macaulay2 system, of the examples mentioned in the works *Eriksson et al.* 2004 and *Sturmfels&Sullivant* 2005. Supervised by Dr Dimitra Kosta.

PROFESSIONAL EXPERIENCE

Lecture Analysis Research Assistant

Jan. 2024 - Jun. 2025

Conducted a data analysis of lecturing habits together with a team of other students, as part of a study into teaching practices conducted by Woods, Kinnear & Galloway. I was responsible for writing R code used in data analysis as well as for data visualisation. Through weekly technical discussions with my team members I strongly improved my scientific communication skills. My work was incorporated by the authors into their official communication.

Maths Outreach Intern

Jun. 2023 - Aug. 2023

Worked closely with Dr Francesca Iezzi on conducting a data analysis on the SIMD metrics of partnered schools as well as on creating new activities which encourage mathematical thinking. Took a proactive role in the research for the new instalment of the Geometry Art & Mind exhibition; my contributions culminated into the development of a new exhibition on Sugihara Shapes which I together with Dr Iezzi showcased at the Talking Maths in Public 2023 conference. Furthermore, I automated the look-up system used to identify the SIMD metrics of Scottish schools, which greatly aided my team's efforts in completing a comprehensive data analysis of the partnered schools.

OUTREACH EXPERIENCE

Member of the Maths Outreach Team at Edinburgh since 2021; highly involved with the numerous outreach initiatives at the University of Edinburgh. The following are selected high-profile projects:

Maths Circles

Oct. 2021 - Jul. 2026

Part of a team which plans and runs maths activities for school pupils every month. This involved working with children on various problems in an interactive manner, probing their understanding and carefully guiding them to a potential solution, developing confidence in their problem-solving abilities. I ran mini-classes on various mathematical topics, including an exposition to topology for a high-school audience.

MathPALS

Jan. 2023 - Jul. 2026

Committee member of the university's maths peer-learning support group: responsible for chairing weekly volunteer meetings and working closely together with the Student Support Team to ensure smooth day-to-day running of the scheme. Furthermore, this role encompassed tutoring responsibilities for first-year mathematics courses.

Maths Walking Tours

Jan. 2023 - Jun. 2026

Responsible for the preparation and running of the annual Edinburgh mathematical walking tour. Improved my oral presentation skills by humorously lecturing on the mathematical history of the city. Through personalised and interactive activities, I delivered a highly-engaging hour-long presentation to a non-expert audience.

Nabój Competition

Jan. 2024 - Jun. 2026

Involved in planning and organising the annual Náboj maths competition with the University of Edinburgh. Worked both in an administrative role prior to the competition as well as an active volunteer on the day, directing the working groups on tasks to complete and proactively addressing problems that might arise, leading to a successful and fun event for the participants.

Edinburgh Science Festival

Jan. 2023 - Apr. 2025

Led the outreach teams' contribution to the Edinburgh Science Festival, working closely with the individual project teams to research and develop the activities, developing my team-working and adaptability skills. Helped coordinate on a larger scale the entire exhibit, ensuring that the whole project is completed on time, and guaranteeing a fun and informative experience for the visitors.